

**MR. COLINGUATICS: A MOBILE GAME-BASED LEARNING
APPLICATION IN TEACHING LOGIC**

Questionnaire for Reliability Testing of the Research Instrument

Name: _____
Gender: _____

Date: _____
Age: _____

Instructions: This questionnaire is part of the data-gathering of the research instrument of the third-year BSEd–Mathematics research on “MR. COLINGUATICS: A Mobile Game-Based Learning Application in Teaching Logic.” Please read each item carefully, do your best to answer honestly, and **ENCIRCLE** the letter of the correct answer for more accurate results. All responses will be used only for research and kept confidential under the *Data Privacy Act of 2012*.

1. Which logical connective is represented by the symbol $\wedge$?
 - a. OR
 - b. AND
 - c. NOT
 - d. If...then
2. In logic, the symbol $\vee$ represents:
 - a. AND
 - b. NOT
 - c. OR
 - d. If and only if
3. The logical symbol $\neg$ represents which of the following?
 - a. NOT
 - b. If... then
 - c. Sigma Notation
 - d. Exclusive Disjunction
4. What does the symbol $\rightarrow$ represent in logic?
 - a. If and only if
 - b. If... then
 - c. AND
 - d. OR
5. The symbol $\leftrightarrow$ is used for which connective?
 - a. If and only if
 - b. If... the
 - c. OR
 - d. AND
6. What logical connective does the symbol $\oplus$ represent?
 - a. Exclusive Disjunction
 - b. NOT
 - c. Union
 - d. Subset
7. In logic, the symbol $\forall$ represents which of the following?
 - a. Union
 - b. There exists
 - c. Subset
 - d. For all
8. Which statement correctly describes the logical symbol $\exists$?
 - a. For all
 - b. Intersection
 - c. Empty set
 - d. There exists
9. What does the symbol $\subseteq$ mean in logic?
 - a. Subset
 - b. Membership Symbol
 - c. Union
 - d. Intersection
10. In mathematical logic, the set symbol $\subset$ represents which of the following?
 - a. Union
 - b. Intersection
 - c. Proper Subset
 - d. Empty Set
11. In writing a formal solution, a student needs to show that two values are not equal. Which notation should the student use?
 - a. $x \equiv y$
 - b. $x \neq y$
 - c. $x \leq y$
 - d. $x \approx y$
12. A researcher wants to indicate in a report that two measured values are close in value but not exactly equal. Which symbol pair should be used?
 - a. $x \propto y$
 - b. $x \geq y$
 - c. $x \equiv y$
 - d. $x \approx y$
13. A student writes, “The value of x does not exceed the value of y .” Which symbolic form should replace this statement?
 - a. $x \leq y$
 - b. $x \neq y$
 - c. $x \propto y$
 - d. $x \equiv y$
14. Let p : “The number is even” and q : “The number is positive.” Which expression should be used to represent the statement “The number is even and positive”?
 - a. $p \vee q$
 - b. $p \rightarrow q$
 - c. $p \wedge q$
 - d. $p \leftrightarrow q$
15. Let p : “A figure is a square” and q : “A figure has four sides.” Which symbolic expression should be used for the statement “If a figure is a square, then it has four sides”?
 - a. $p \rightarrow q$
 - b. $p \wedge q$
 - c. $p \oplus q$
 - d. $p \vee q$
16. Let p : “The gate is open” and q : “Visitors may enter.” Which expression should be used to represent the statement “The gate is open if and only if visitors may enter”?
 - a. $p \wedge q$
 - b. $p \vee q$
 - c. $p \rightarrow q$
 - d. $p \leftrightarrow q$
17. A student is asked to select a symbolic statement that can already be assigned a truth value. Which of the following should the student choose?
 - a. $x \neq y$
 - b. $\forall x (x=x)$

- c. Open the door. d. Let x be a number.
18. During a lesson on logical statements, a student is asked to exclude the expression that cannot yet be assigned a definite truth value unless its variables are specified or quantified. Which expression should be excluded?
- a. $p \wedge q$ c. $x \geq y$
b. $\forall x (x = x)$ d. $p \leftrightarrow q$
19. Let p : "Ana will present the report" and q : "Ben will present the report." The teacher wants to express that only one of the two will present. Which symbolic form should be used?
- a. $p \oplus q$ c. $p \leftrightarrow q$
b. $p \wedge q$ d. $p \vee q$
20. A mathematics teacher wants to symbolize the statement "For every value of x , the rule applies." Which symbol should be used to begin the statement?
- a. $\exists x$ c. $x \equiv y$
b. $\neg x$ d. $\forall x$
21. A student translates "It is raining and I am late" as $p \vee q$. What error is present in this translation?
- a. Misuse of negation.
b. Incorrect connective used for conjunction.
c. Misinterpretation of variables.
d. Wrong quantifier.
22. Consider the statement $\neg(p \wedge q)$. Which analysis correctly explains its meaning?
- a. Both p and q are true.
b. Either p or q is false (or both).
c. Only p is false.
d. Both are always false.
23. A student says that $\neg p$ is true whenever p is true. What is the error in this reasoning?
- a. Negation reverses the truth value of p .
b. Negation keeps the same truth value.
c. Negation applies only to variables.
d. Negation is optional.
24. Which situation makes $p \oplus q$ TRUE?
- a. Both p and q are true.
b. Both p and q are false.
c. Exactly one of p or q is true.
d. p implies q
25. Which comparison correctly analyzes $p \rightarrow q$ and $q \rightarrow p$?
- a. They are always equivalent.
b. They have different meanings and truth conditions.
c. Both are always true.
d. Both are always false.
26. Which statement correctly combines negation and conjunction?
- a. $\neg p \wedge q$ means both p and q are true
b. $\neg p \wedge q$ means p is false and q is true
c. $\neg p \wedge q$ means both are false
d. $\neg p \wedge q$ means p implies q
27. Which statement best describes $p \leftrightarrow q$?
- a. p is true only if q is true
b. q is true only if p is true
c. p and q must always have identical truth values
d. Either p or q must be true
28. Let the statement be: $\neg(p \rightarrow q) \wedge r$. Which interpretation best describes when this statement is TRUE?
- a. p is false, q is true, and r is true
b. p is true, q is false, and r is true
c. p and q are both true, and r is false
d. p is false, q is false, and r is true
29. Which statement best describes the difference between:
- $\forall x P(x)$
 $\exists x P(x)$
- a. Universal requires all, existential requires at least one.
b. Both require all elements to satisfy $p(x)$.
c. Existential requires all, universal requires none.
d. Both are always false.
30. A student interprets: $p \wedge (q \vee r)$ as $(p \wedge q) \vee r$. What is the correct analysis?
- a. Distribution was applied incorrectly.
b. Commutative property was applied correctly.
c. A negation rule was used.
d. Biconditioning was misused.

